Bihar Engineering University, Patna

B.Tech. 1st Semester Examination, 2023

Course: B.Tech. *Time: 03 Hours*

Code: 101101 *Subject: Physics (Mechanics & Mechanics of Solid)* *Full Marks: 70*

Instructions:-

(i) The marks are indicated in the right-hand margin.
(ii) There are NINE questions in this paper.
(iii) Attempt FIVE questions in all.
(iv) Question No. 1 is compulsory.
(v) Symbols used (if any) have their usual meanings

Q.1 *Choose the correct answer of the following (Any seven question only):* **[2 x 7 = 14]**

(a) The moment of inertia of an arbitrarily shaped rigid body is a
(i) Scalar (ii) Vector (iii) Tensor (iv) Dimensionless number

(b) The curl of a gradient is
(i) Zero (ii) Non-zero (iii) Undefined (iv) Infinite

(c) In a lightly damped oscillator the frequency
(i) Depends on the square of amplitude (ii) Does not depend on the amplitude
(iii) Depends on the damping constant (iv) Does not depend on the damping

(d) As opposed to a standing bicycle a moving bicycle does not fall easily because
(i) The cyclist keeps moving slightly left and right to maintain the C.G in the plane of the bicycle
(ii) The cycle is balanced by air resistance
(iii) The angular momentum in the wheels stabilizes the cycle
(iv) The cyclist leans forward to lower the CG

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(e) The Foucault pendulum undergoes precession because of
(i) Gravitational force of the earth
(ii) Coriolis force on the earth
(iii) Magnetic field of the earth
(iv) The orbiting of the moon around the earth

(f) Which of the following is NOT Kepler's law
(i) Each planet moves in an ellipse with the Sun at one focus
(ii) Planet's period of revolution T is related to the major axis of the ellipse A by $T^2 = k A^3$
(iii) Planet's period of revolution T is related to the major axis of the ellipse A by $T^3 = k A^2$
(iv) The radius vector from the Sun to the planet sweeps equal areas in equal times.

(g) Which of the following is not true for conservative force
(i) Curl of the force is zero
(ii) Gravitational force is conservative in nature
(iii) Work done is independent of the path
(iv) Curl of the force is non zero

(h) The Newton's law gives the relationship between force, mass and
(i) Time (ii) Velocity
(iii) Acceleration (iv) Impulse

(i) Velocity or energy resonance in the forced oscillator take place when
(i) $\omega = \sqrt{\omega_0^2 - 2b^2}$ (ii) $\omega = \omega_0$
(iii) $\omega = 2b$ (iv) $\omega = 2\omega_0$
Where ω is frequency of force, ω_0 is natural frequency of the oscillator and $2b = \frac{\beta}{m}$ is damping coefficient

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(j) A spinning top is an example of
(i) Two dimensional rigid body motion
(ii) Liner motion
(iii) Three dimensional rigid body motion
(iv) One dimensional rigid body motion

Q.2 (a) State the definition of scalar and vector quantities. Explain how they transform under rotation. [5]

(b) Under an anti-clockwise rotation by an angle θ of a *2d* Cartesian coordinate system find out the transformation rules for the components of an arbitrary vector $\vec{A}$. Show that the scalar product $\vec{A}.\vec{B}$ is invariant under rotation of the coordinate system. [9]

Q.3 (a) Define inertial and non-inertial frames. Explain the meaning of fictitious force [5]

(b) Derive an expression for acceleration in a rotating frame and obtain the centripetal and Coriolis terms in the acceleration. Explain briefly how the Coriolis forces are useful in explaining the weather system [9]

Q.4 Write short technical notes on: [7 x 2 = 14]

(a) Satellite manoeuvre

(b) Foucault pendulum

Q.5 (a) Show that the conservative force acting on a particle is always perpendicular to the equipotential surface. [5]

(b) Prove that the work done by a force which is the gradient of a potential ($\vec{F} = -\vec{\nabla} U$) is path independent. [4]

(c) The potential function is given as $U = -2x - z^2$. Find the work done by the resulting force in moving a particle from the origin to a point (2,2,2) along a straight line. [5]

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Q.6 (a) Explain overdamped, underdamped and critically damped harmonic motion. [7]

(b) Write an equation for a forced harmonic oscillator and obtain the conditions of resonance. [7]

Q.7 (a) Derive the moment of inertia tensor of a rigid body and explain the concept of principal axes. [7]

(b) A uniform solid disc of mass M and radius R having moment of inertia $MR^2/2$ about an axis passing through its centre has initial uniform angular velocity. A constant friction force of magnitude f is then applied on the rim in tangential direction so that the disc finally comes to a stop in time T. Find the magnitude of the friction force. [7]

Q.8 (a) Derive the conditions for elliptic, hyperbolic and parabolic orbits in central force motion. [8]

(b) Show that the total angular momentum of a solid body can be written as the sum of the angular momentum of the centre of mass and the angular momentum about the centre of mass [6]

Q.9 (a) Derive Euler's equations for the rotational motion of a rigid body and explain its importance. [8]

(b) Prove that the angular momentum is conserved in a central force motion and that the orbit of the particle is always planar. [6]

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